Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — qualifier level (Listas 3–4)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption, present bias and interest rates

(Kurlat chs. 6, 9; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives 40 periods ($t = 0, \ldots, 39$) with CRRA utility, $\sigma = 2$, $\beta = 0.97$, faces $r = 4\%$ and has total wealth W (financial plus human, in present value at $t = 0$). The share of W consumed at $t = 0$ is:

A. 3.42%

B. 4.55%

C. 2.50%

D. 3.85%

Answer: B

Step — path. Euler: $c_t = c_0[\beta(1+r)]^{t/\sigma}$; budget $\sum_{t=0}^{39} c_t(1+r)^{-t} = W$ gives $c_0 \sum_{t=0}^{39} \gamma^t = W$ with $\gamma \equiv \beta^{1/\sigma}(1+r)^{1/\sigma-1} = \sqrt{0.97/1.04} = 0.96576$.

Step — share. $c_0/W = \frac{1-\gamma}{1-\gamma^{40}} = \frac{0.03424}{0.75182} = 4.55\%$. A is the infinite-horizon $1-\gamma$; C is $1/40$; D is $r/(1+r)$.

Ref: Kurlat (2020), §6.3, eqs. (6.3.6)–(6.3.9), pp. 117–120.

b) (3 points) *Consumption × general equilibrium.* A two-period endowment economy with a representative household: $y_2/y_1 = 1.03$, $\beta = 0.97$, CRRA with $\sigma = 2$. The equilibrium real interest rate is:

A. 3.09%

B. 6.19%

C. 9.37%

D. 3.00%

Answer: C

Step — clearing. No one can lend in the aggregate: $c_t = y_t$. Euler: $y_1^{-\sigma} = \beta(1+r)y_2^{-\sigma} \Rightarrow 1 + r = (y_2/y_1)^{\sigma}/\beta = 1.0609/0.97$, $r = 9.37\%$. B uses $\sigma = 1$; A ignores growth.

Ref: Kurlat (2020), §9.1, pp. 165–168.

c) (2 points, T or F) *Consumption × credit markets.* A household strictly inside the region where it neither borrows nor lends (borrowing rate above the deposit rate) starts saving after a small rise in the deposit rate.

Answer: F

Step — kink. At the kink the MRS $u'(y_1)/[\beta u'(y_2)]$ lies strictly between $1 + r_s$ and $1 + r_b$; a small rise in r_s keeps it above $1 + r_s$, so saving is still unattractive and consumption does not change.

Ref: Kurlat (2020), Exercise 6.6 (kinked budget sets).

d) (2 points, T or F) *Consumption × fiscal policy.* A household at a binding borrowing limit receives a lump-sum tax cut of 100 today, financed by a tax increase of $100(1 + r)$ next period. If the limit still binds after the cut, current consumption rises by exactly 100.

Answer: T

Step — why. With the limit binding, $c_1 = y_1 - \tau_1 + b$: consumption equals current resources, so it moves one for one with the tax cut; Ricardian equivalence fails because the household cannot borrow against the future tax.

Ref: Kurlat (2020), §6.2 and Exercise 6.5.

Question 2. (15 points) Discursive. *Consumption × behavioral economics × general equilibrium.* Three periods $t = 0, 1, 2$; in (a)–(c) the interest rate is zero and the household has wealth W at date 0. The self at date t maximizes $u(c_t) + \beta \sum_{s>t} u(c_s)$ with $u(c) = c^{1-\sigma}/(1-\sigma)$ ($\ln c$ if $\sigma = 1$) and $0 < \beta < 1$.

a) (3 points) Find the plan chosen with commitment (self 0 chooses all three consumptions), by a naive self 0 (who believes future selves will follow its plan), and by a sophisticated self 0 (who anticipates self 1's choice).

Step — self 1. $\max u(c_1) + \beta u(c_2)$ s.t. $c_1 + c_2 = W_1$: $c_1^{-\sigma} = \beta c_2^{-\sigma} \Rightarrow c_2 = \beta^{1/\sigma} c_1$, so $c_1 = kW_1$, $c_2 = (1-k)W_1$, $k \equiv \frac{1}{1+\beta^{1/\sigma}} > \frac{1}{2}$.

Step — commitment. Self 0 weighs dates 1 and 2 equally: $c_1 = c_2 = c$, and $u'(c_0) = \beta u'(c) \Rightarrow c_0 = \beta^{-1/\sigma} c$; the budget gives $c = \frac{W}{2+\beta^{-1/\sigma}}$.

Step — naive. Believing $c_1 = c_2$, self 0 picks the commitment c_0; then self 1 re-optimizes: $c_1 = kW_1$, $c_2 = (1-k)W_1$.

Step — sophisticated. Self 0 solves $\max_{c_0} u(c_0) + \beta[u(kW_1) + u((1-k)W_1)]$, $W_1 = W - c_0$. FOC: $c_0^{-\sigma} = \beta[k^{1-\sigma} + (1-k)^{1-\sigma}]W_1^{-\sigma}$, so $c_0 = \frac{x_s}{1+x_s} W$ with $x_s \equiv \left(\beta[k^{1-\sigma} + (1-k)^{1-\sigma}]\right)^{-1/\sigma}$ (for $\sigma = 1$, $x_s = 1/(2\beta)$).

Ref: Romer (2012), §8.6 and problem 8.15, pp. 397–404; Kurlat (2020), §6.4.

Grading rubric.

- Self 1's rule: 0.5 pt.
- Commitment plan: 1 pt.
- Naive plan (two steps): 0.5 pt.
- Sophisticated FOC and c_0: 1 pt.

b) (3 points) Determine when the sophisticated self 0 consumes more, less or the same as the naive one, and explain the mechanism.

Step — compare. Naive: $c_0/W_1 = x_n = \beta^{-1/\sigma}/2$, so $x_n^{-\sigma} = \beta 2^{\sigma} = 2\beta\varphi(\frac{1}{2})$ with $\varphi(z) \equiv z^{1-\sigma}$. Sophisticated: $x_s^{-\sigma} = \beta[\varphi(k) + \varphi(1-k)]$. Since $z \mapsto z^{-\sigma}$ is decreasing and $c_0 = \frac{x}{1+x} W$ rises with x: $c_0^s < c_0^n \iff \varphi(k) + \varphi(1-k) > 2\varphi(\frac{1}{2})$.

Step — Jensen. $\varphi''(z) = \sigma(\sigma-1)z^{-\sigma-1}$: φ is strictly convex if $\sigma > 1$, linear in logs at $\sigma = 1$ (both sides equal), concave if $\sigma < 1$. With $k \neq \frac{1}{2}$: $\sigma > 1 \Rightarrow c_0^s < c_0^n$; $\sigma = 1 \Rightarrow$ equal; $\sigma < 1 \Rightarrow c_0^s > c_0^n$.

Step — mechanism. The sophisticate knows the wealth it leaves will be split unevenly. With high curvature ($\sigma > 1$) an uneven split raises the marginal value of extra wealth, so it saves more to compensate; with $\sigma < 1$ the misallocation lowers that value and it consumes more now. Under log the two forces cancel exactly.

Ref: Romer (2012), problem 8.15(b)(ii).

Grading rubric.

- Reduction to the comparison of φ values: 1.5 pts.
- Jensen argument with all three cases: 1 pt.
- Mechanism: 0.5 pt.

c) (3 points) What share of W would self 0 pay for an illiquid account that commits future consumption? Find a formula, its limits as $\beta \to 1$ and as $\sigma \to \infty$, and compute it for $\sigma = 2$, $\beta = 1/2$.

Step — homogeneity. For $\sigma \neq 1$ every plan is proportional to W, so self 0's value is $V_j(W) = W^{1-\sigma} v_j$ (j = commitment c, sophisticated s). The willingness to pay x solves $V_c((1-x)W) = V_s(W)$:

$$x = 1 - \left(\frac{v_s}{v_c}\right)^{\frac{1}{1-\sigma}}; \qquad \sigma = 1: \; x = 1 - \exp\Big(-\frac{V_c - V_s}{1+2\beta}\Big).$$

It does not depend on W.

Step — limits. $\beta \to 1$: $k \to \frac{1}{2}$, all plans coincide, $x \to 0$. $\sigma \to \infty$: $\beta^{1/\sigma} \to 1$, $k \to \frac{1}{2}$, self 1 also splits evenly, $x \to 0$ (the table in (e) shows the decline).

Step — numbers. $\sigma = 2$, $\beta = \frac{1}{2}$, $W = 100$, plans (c_0, c_1, c_2):

commitment (41.42, 29.29, 29.29), naive (41.42, 34.31, 24.26),
sophisticated (41.06, 34.53, 24.41);

$V_c = -0.05828$, $V_s = -0.05932$, $V_n = -0.05932$; $x = 1 - V_c/V_s = 1.74\%$ (log: 2.90%).

Ref: Romer (2012), problem 8.15; Kurlat (2020), §6.4 (commitment, defaults).

Grading rubric.

- Homogeneity and the formula: 1.5 pts.
- Limits: 0.5 pt.
- Numbers: 1 pt.

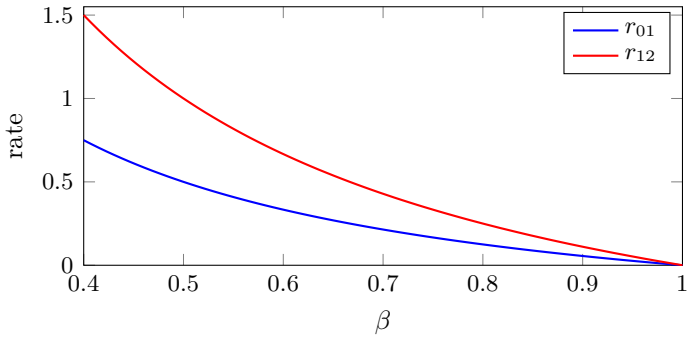
d) (3 points) Now every household receives y at each date and households are identical and sophisticated; they can borrow and lend at r_{01} (between dates 0 and 1) and r_{12}. Find the equilibrium interest rates, show whether they depend on σ, and draw them against $\beta \in [0.4, 1]$.

Step — clearing. Identical households cannot lend to each other: $c_t = y$ at every date.

Step — r_{12}. Self 1's Euler at $c_1 = c_2 = y$: $y^{-\sigma} = \beta(1+r_{12})y^{-\sigma} \Rightarrow 1 + r_{12} = 1/\beta$.

Step — self 1's rule at these prices. With $\beta(1+r_{12}) = 1$, self 1 sets $c_2 = c_1$ and $c_1 = W_1/(1+\beta)$, where $W_1 = y + y/(1+r_{12}) = y(1+\beta)$ in equilibrium: $c_1 = y$. Self 0's value of W_1 is $2u(W_1/(1+\beta))$, with marginal value $\frac{2}{1+\beta}u'(y)$.

Step — r_{01}. Self 0's Euler: $u'(y) = (1+r_{01})\beta\frac{2}{1+\beta}u'(y) \Rightarrow 1 + r_{01} = \frac{1+\beta}{2\beta}$. Neither rate depends on σ (equal endowments). At $\beta = \frac{1}{2}$: $r_{01} = 50\%$, $r_{12} = 100\%$; with $\beta = 1$ both are zero.



Economic intuition. Present bias makes everyone want to bring consumption forward; with nobody to borrow from, rates rise until each self is content with its endowment, and the near-term rate rises most.

Ref: Kurlat (2020), §9.1 (endowment equilibrium), pp. 165–168.

Grading rubric.

- Market clearing and r_{12}: 1 pt.
- Self 0's marginal value of wealth and r_{01}: 1 pt.
- Graph: axes, two decreasing curves meeting at zero when $\beta = 1$, r_{12} above r_{01}: 1 pt.

e) (3 points) *Parameter analysis.* How do the value of commitment and the equilibrium rates depend on β and σ? Prove the signs you can, fill the value of commitment for $\sigma \in \{1, 1.5, 2, 5\}$ and $\beta \in \{0.5, 0.7, 0.9\}$, and read the results.

Step — rates. $\partial r_{01}/\partial\beta = -\frac{1}{2\beta^2} < 0$, $\partial r_{12}/\partial\beta = -\frac{1}{\beta^2} < 0$, $\partial/\partial\sigma = 0$; $r_{12} > r_{01}$ for $\beta < 1$.

Step — table. Willingness to pay for commitment (% of W):

	$\beta = 0.5$	$\beta = 0.7$	$\beta = 0.9$
$\sigma = 1$	2.90	0.92	0.09
$\sigma = 1.5$	2.20	0.65	0.06
$\sigma = 2$	1.74	0.50	0.05
$\sigma = 5$	0.75	0.21	0.02

It falls with β (less disagreement between selves) and with σ (self 1's split approaches the even split).

Step — reading. The demand for commitment is concentrated where present bias is strong and curvature is low: that is where defaults and illiquid accounts (401(k)s, Brazil's FGTS) matter. In equilibrium the same bias shows up as high short-term real rates, independent of σ when endowments are flat.

Ref: Kurlat (2020), §6.4 and §9.1; Romer (2012), problem 8.15.

Grading rubric.

- Signs of the rates: 1 pt.
- Table: 1 pt.
- Reading of both dependences: 1 pt.

Block 2 — Labor supply, taxes, transfers and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Labor × consumption smoothing.* Per-period utility $\ln c - \psi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}$, $\varepsilon = 0.5$; 40 periods with $\beta(1+r) = 1$ and equal wages. The wage rises 10% in one period only. To first order, hours in that period rise by:

A. 4.9%

B. 5.0%

C. 0%

D. 3.3%

Answer: A

Step — FOCs. $\psi h_t^{1/\varepsilon} = \lambda w_t$ with $\lambda = 1/c$ constant across periods; so $d\ln h_1 = \varepsilon(d\ln w_1 - d\ln c)$ and $d\ln h_t = -\varepsilon\, d\ln c$ for $t \neq 1$.

Step — budget. $d\ln c = \sum_t s_t(d\ln w_t + d\ln h_t)$ with earnings shares $s_t = 1/40$; solving, $d\ln c = s_1 d\ln w_1$ and $d\ln h_1 = \varepsilon(1-s_1)d\ln w_1 = 0.5 \cdot 0.975 \cdot 10\% = 4.9\%$ (4.6% for the exact discrete change). B is the pure Frisch response; C the response to a permanent raise.

Ref: Kurlat (2020), §7.4, pp. 140–141.

b) (3 points) *Search × wage setting.* In the steady-state search model with $f = \mu\theta^\alpha$, $\alpha = 0.5$, free entry $(y-w)q(\theta) = \kappa(r+s)$, the wage is rigid at $w = 0.95$ while productivity $y = 1$ moves. The elasticity of tightness θ with respect to y is:

A. 3.7

B. 40

C. 20

D. 2.0

Answer: B

Step — log-differentiate. $q = \mu\theta^{\alpha-1}$: $\ln(y-w) + (\alpha-1)\ln\theta = \text{const}$, so $\frac{d\ln\theta}{d\ln y} = \frac{1}{1-\alpha}\frac{y}{y-w} = 2 \cdot 20 = 40$. A is the bargained-wage value: rigid wages are one answer to Shimer's puzzle.

Ref: Kurlat (2020), §7.5, p. 146; Romer (2012), problem 10.15.

c) (2 points, T or F) *Search × efficiency.* With a general constant-returns matching function, the search equilibrium is efficient iff the workers' share of the surplus equals the elasticity of matches with respect to unemployment, evaluated at the equilibrium tightness.

Answer: T

Step — Hosios. The planner's and the market's conditions coincide iff the firms' share equals the vacancy elasticity of matches; with constant returns the two elasticities add to one, and both are evaluated where the economy is.

Ref: Romer (2012), §10.7 and problem 10.16.

d) (2 points, T or F) *Labor × transfers.* With $u = \ln c + \alpha\ln(1-h)$ and a lump-sum transfer $T > 0$ that each household takes as given, an individual household's hours do not respond to its after-tax wage.

Answer: F

Step — supply. $h = \frac{1}{1+\alpha} - \frac{\alpha T}{(1+\alpha)w_n}$ rises with w_n when $T > 0$; hours are flat in w_n only when $T = 0$ (Question 4(a)).

Ref: Kurlat (2020), §7.2.

Question 4. (15 points) Discursive. *Labor supply × taxes and transfers × welfare.* Households have $u = \ln c + \alpha\ln(1-h)$ with one unit of time; a linear technology pays $w = 1$ per hour; labor income is taxed at τ and the revenue is returned as a lump-sum transfer T that each household takes as given. Baseline $\alpha = 0.8$.

a) (3 points) Derive individual labor supply given (τ, T). Show what the tax does when $T = 0$, and find equilibrium hours $h(\tau)$ when the budget balances.

Step — FOC. $\max \ln[(1-\tau)h + T] + \alpha\ln(1-h)$: $\frac{1-\tau}{c} = \frac{\alpha}{1-h}$, i.e. $\alpha c = (1-\tau)(1-h)$ with $c = (1-\tau)h + T$.

Step — individual supply. $\alpha[(1-\tau)h + T] = (1-\tau)(1-h) \Rightarrow h = \frac{1}{1+\alpha} - \frac{\alpha T}{(1+\alpha)(1-\tau)}$. With $T = 0$, $h = \frac{1}{1+\alpha}$: the tax's income and substitution effects cancel.

Step — balanced budget. $T = \tau h$ (each household still takes T as given): $c = h$, and the FOC gives $\frac{1-\tau}{h} = \frac{\alpha}{1-h} \Rightarrow h(\tau) = \frac{1-\tau}{1-\tau+\alpha}$, decreasing in τ; $h(0) = 0.556$.

Ref: Kurlat (2020), §7.2–7.3 and Exercise 7.5, pp. 131–140, 148–149.

Grading rubric.

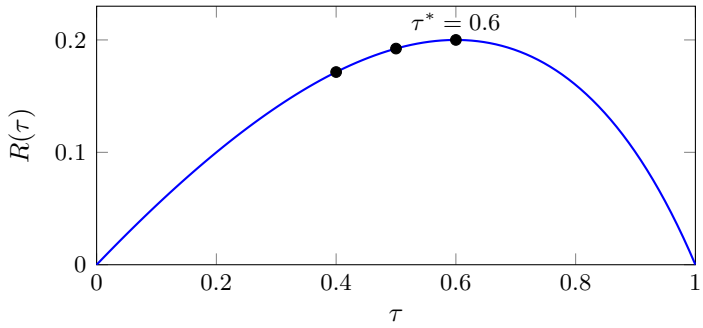
- FOC: 1 pt.
- Individual supply and the $T = 0$ case: 1 pt.
- Equilibrium $h(\tau)$ with T substituted after the FOC: 1 pt. Substituting before: max. 1.5 pts in the item.

b) (3 points) Find the tax rate that maximizes revenue (the size of the transfer program), prove that revenue is single-peaked, and draw revenue against τ.

Step — revenue. $R(\tau) = \tau h(\tau) = \frac{\tau(1-\tau)}{1+\alpha-\tau}$. The numerator of R' is $N(\tau) = \tau^2 - 2(1+\alpha)\tau + (1+\alpha)$.

Step — single peak. N is a convex parabola with $N(0) = 1+\alpha > 0$ and $N(1) = -\alpha < 0$: exactly one root in $(0,1)$, the other above 1. So $R' > 0$ before and $R' < 0$ after

$$\tau^* = (1+\alpha) - \sqrt{\alpha(1+\alpha)} = 1.8 - 1.2 = 0.6, \qquad R(\tau^*) = 0.6 \cdot \tfrac{0.4}{1.2} = 0.2.$$



Ref: Kurlat (2020), §7.3.

Grading rubric.

- Revenue and the sign of R': 1 pt.
- Single-peak proof and τ^*: 1 pt.
- Graph: axes, hump-shaped curve, peak at $(0.6, 0.2)$: 1 pt.

c) (3 points) Derive the welfare cost of the program in consumption-equivalent terms (the share of consumption at $\tau = 0$ the household would give up to avoid the program), prove that it is zero to first order at $\tau = 0$ and increasing, and compute it at $\tau = 0.4$, 0.5, 0.6.

Step — welfare. With $c = h$: $U(\tau) = \ln h(\tau) + \alpha \ln(1 - h(\tau))$. The cost x solves $U(0) + \ln(1 - x) = U(\tau)$: $x(\tau) = 1 - e^{U(\tau) - U(0)}$.

Step — shape. $U'(\tau) = h'(\tau)\left[\frac{1}{h} - \frac{\alpha}{1-h}\right]$. At $\tau = 0$, $h = \frac{1}{1+\alpha}$ and the bracket is zero: no first-order cost. For $\tau > 0$, $h < \frac{1}{1+\alpha}$, so $\frac{1}{h} > 1 + \alpha > \frac{\alpha}{1-h}$; with $h' < 0$, $U' < 0$: the cost rises with τ.

Step — numbers. $\tau = 0.4$: $h = 0.429$, $x = 5.68\%$; $\tau = 0.5$: $h = 0.385$, $x = 10.18\%$; $\tau = 0.6$: $h = 0.333$, $x = 17.01\%$.

Economic intuition. The program moves nothing between households; its whole effect is a wedge on the price of leisure, and the loss grows with the square of the wedge near zero.

Ref: Kurlat (2020), Exercise 7.5 (welfare comparison).

Grading rubric.

- Welfare measure: 1 pt.
- First-order zero and monotonicity proved: 1 pt.
- Numbers: 1 pt.

d) (3 points) At $\tau = 0.4$, compute the Marshallian and Hicksian elasticities of individual hours with respect to the after-tax wage, the balanced-budget elasticity $d \ln h / d \ln(1 - \tau)$, and the Frisch elasticity. Which one does Prescott's experiment use?

Step — Marshallian. From (a), $\partial h / \partial w_n = \frac{\alpha T}{(1+\alpha) w_n^2}$, so $e_M = \frac{\alpha T}{(1+\alpha) w_n h} = \frac{\alpha \tau}{(1+\alpha)(1-\tau)} = 0.296$ (using $T = \tau h$).

Step — Hicksian. Slutsky: $\partial h^c / \partial w_n = \partial h / \partial w_n - h \, \partial h / \partial T = \frac{\alpha c}{(1+\alpha) w_n^2}$, so $e_H = \frac{\alpha}{(1+\alpha)(1-\tau)} = 0.741$.

Step — balanced budget and Frisch. $\frac{d \ln h}{d \ln(1-\tau)} = 1 - h = 0.571$. Frisch (marginal utility of wealth fixed): $\frac{\alpha}{1-h} = \lambda w_n$ gives $\frac{1-h}{h} = 1.333$.

Step — which one. Prescott compares economies whose revenue is rebated, so the relevant response is the balanced-budget one: larger than the Marshallian (the transfer offsets part of the income effect) and smaller than the Hicksian.

Ref: Kurlat (2020), §7.2–7.3 and footnote 8, p. 149.

Grading rubric.

- Each elasticity derived: 0.5 pt each (2 pts).
- Which one and why: 1 pt.

e) (3 points) *Parameter analysis.* How do τ^*, hours at $\tau = 0.5$ and the welfare cost at $\tau = 0.5$ depend on α? Prove the signs you can, give the limits of τ^*, fill the table for $\alpha \in \{0.4, 0.8, 1.54, 3\}$ (1.54 is the value in Kurlat's Exercise 7.5) and read it.

Step — signs. $\frac{d\tau^*}{d\alpha} = 1 - \frac{1+2\alpha}{2\sqrt{\alpha(1+\alpha)}} < 0$ because $(1+2\alpha)^2 = 4\alpha(1+\alpha)+1$. Hours $\frac{1-\tau}{1-\tau+\alpha}$ fall with α.

Step — limits. $\alpha \to 0$: $\tau^* \to 1$ (leisure worthless, no distortion until everything is taxed). $\alpha \to \infty$: $\tau^* \to \frac{1}{2}$, from $\sqrt{\alpha^2+\alpha} = \alpha + \frac{1}{2} - \frac{1}{8\alpha} + \dots$

Step — table.

	τ^*	$h(0.5)$	cost at $\tau = 0.5$
$\alpha = 0.4$	0.652	0.556	7.19%
$\alpha = 0.8$	0.600	0.385	10.18%
$\alpha = 1.54$	0.562	0.245	12.75%
$\alpha = 3$	0.536	0.143	14.70%

Step — reading. A higher weight on leisure makes hours more responsive at the margin: the Laffer peak arrives earlier and the same 50% wedge costs more. Prescott's account of Europe needs a large α, i.e. an elasticity at the top of the micro estimates (Kurlat §7.3).

Ref: Kurlat (2020), §7.3 and Exercise 7.5.

Grading rubric.

- Sign of $d\tau^*/d\alpha$ proved and both limits: 1.5 pts.
- Table: 1 pt.
- Reading: 0.5 pt.